



What do the world's largest simulations of cold phase interstellar medium turbulence tell us?

Luncheon Talk, Harvard University

James Beattie

Postdoc. Research Associate / Fellow
Princeton / CITA

In collaboration: Amitava Bhattacharjee (Princeton), Christoph Federrath (ANU),
Ralf Klessen (UH) & Salvatore Cielo (LRZ)

Beattie+2025 (*Nature Astronomy*). The spectrum of magnetized turbulence in the interstellar medium

Beattie & A. Bhattacharjee (submitted *PRL*). Scale-dependent alignment in compressible magnetohydrodynamic turbulence

The cold ISM: A supersonic laboratory for interesting nonlinear physics

- $\sim 20\%$ MW ISM gas is molecular hydrogen, organised into MCs (low-volume filling $\sim 1\text{-}2\%$).
- cold, $T \sim 10\text{ K}$, c_s is low (and $t_{\text{cool}} \ll t_{\text{nl}}$), approximately isothermal
- $\sigma_v/c_s = \mathcal{M} \sim 10$, supersonic (compressible), $\text{Re} \sim 10^9$
- $L \sim 10\text{ pc} \implies \tau = L/\sigma_v \sim \mathcal{O}(\text{Myr})$
- $n \sim 10^3 - 10^{7+}$, huge density contrasts.
- **weakly bounded (not virialised) by their own self-gravity**
 $\alpha_{\text{vir}} = 2 |E_{\text{kin}}| / |E_{\text{grav}}| > 2.$
- threaded by dynamically important B fields, Ohmic
 $\text{Rm} \sim 10^{16}$.

Krumholz & McKee (2005); Federrath & Klessen (2012);
Tritsis+(2016); Xu & Lazarian (2016); Soler & Hennebelle (2017);
Squire & Hopkins (2017); Mocz & Burkart (2018), Burkart & Mocz
(2018); Heyer+(2012); Tritsis+(2018); Hu+(2019); Heyer+(2020);
Krumholz+(2020); Beattie & Federrath (2020); Beattie+(2020);
Körtgen & Soler (2020); Skalidis & Tassis (2020); Federrath+(2021);
Burkhart (2021); Barreto-Mota+(2021); Hopkins+(2022);
Beattie+(2022); Fielding+(2022); Sampson+ (2022); Kriel+(2022);
Galishnikova+(2022); Beattie+(2023)

The cold ISM: A supersonic laboratory for interesting nonlinear physics

- ~ 20% MW ISM gas is molecular hydrogen, organised into MCs (low-volume filling ~ 1-2%).

- cold, $T \sim 10\text{ K}$, c_s is low
isothermal

- $\sigma_v/c_s = \mathcal{M} \sim 10$, **supersonic**

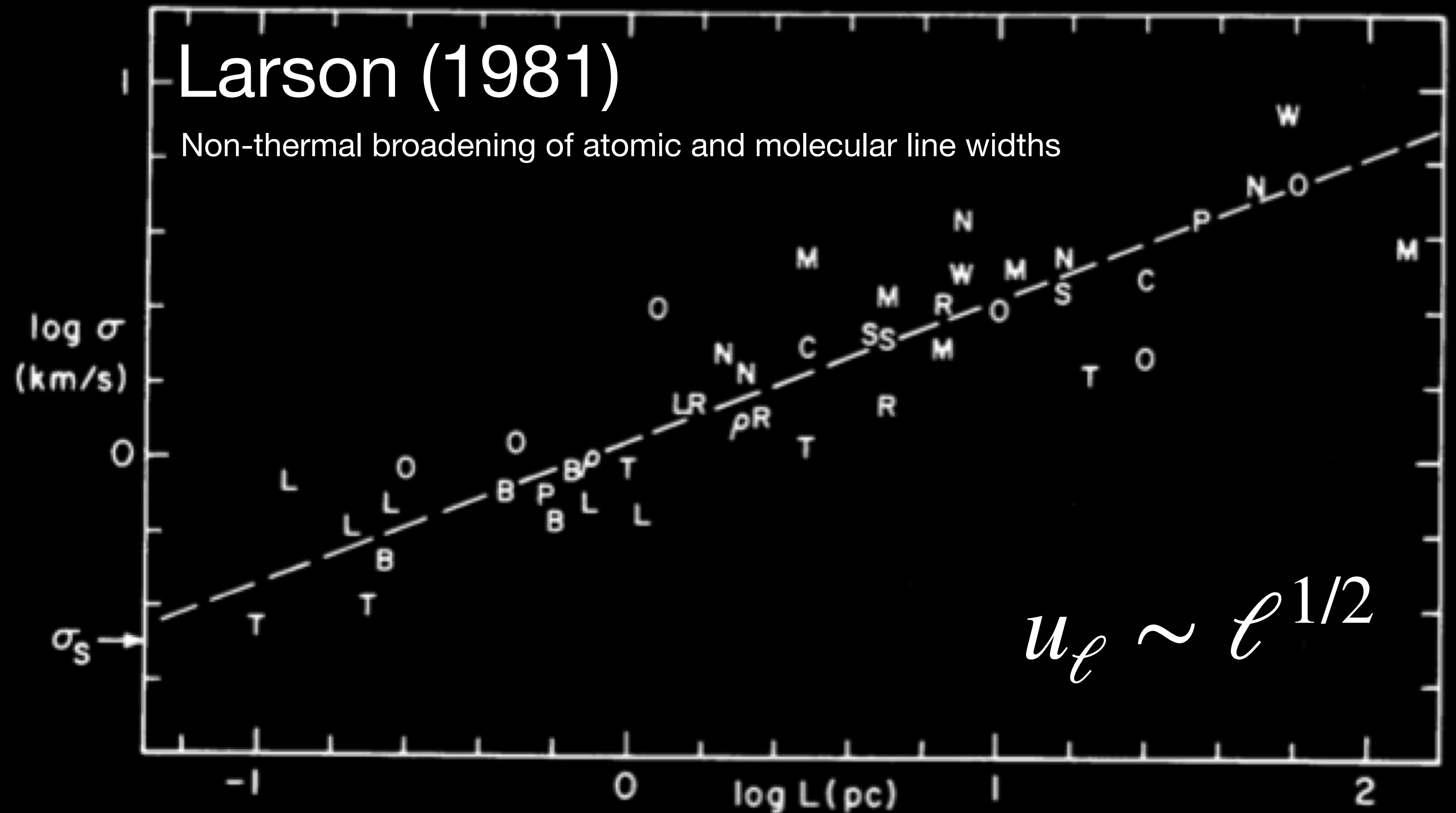
- $L \sim 10\text{ pc} \Rightarrow \tau =$

- $n \sim 10^3 - 10^7+$, huge

- **weakly bounded** (not

$$\alpha_{\text{vir}} = 2 |E_{\text{kin}}| / |E_{\text{grav}}|$$

- threaded by **dynamic pressure**
 $R_m \sim 10^{16}$.

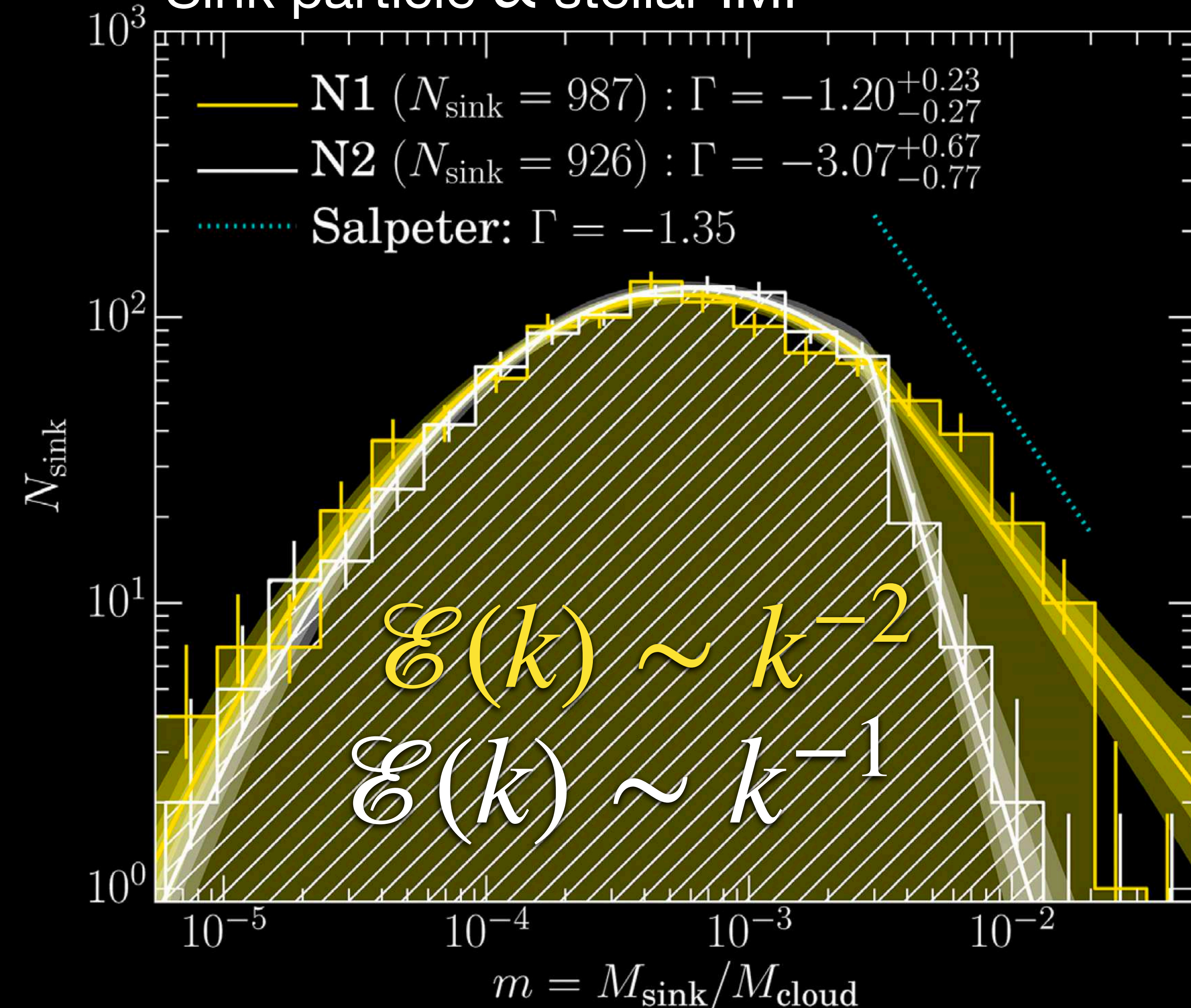


(2012);
 le (2017);
 iart & Mocz
 r+(2020);
 +(2020);
 ath+(2021);
 022);
 el+(2022);

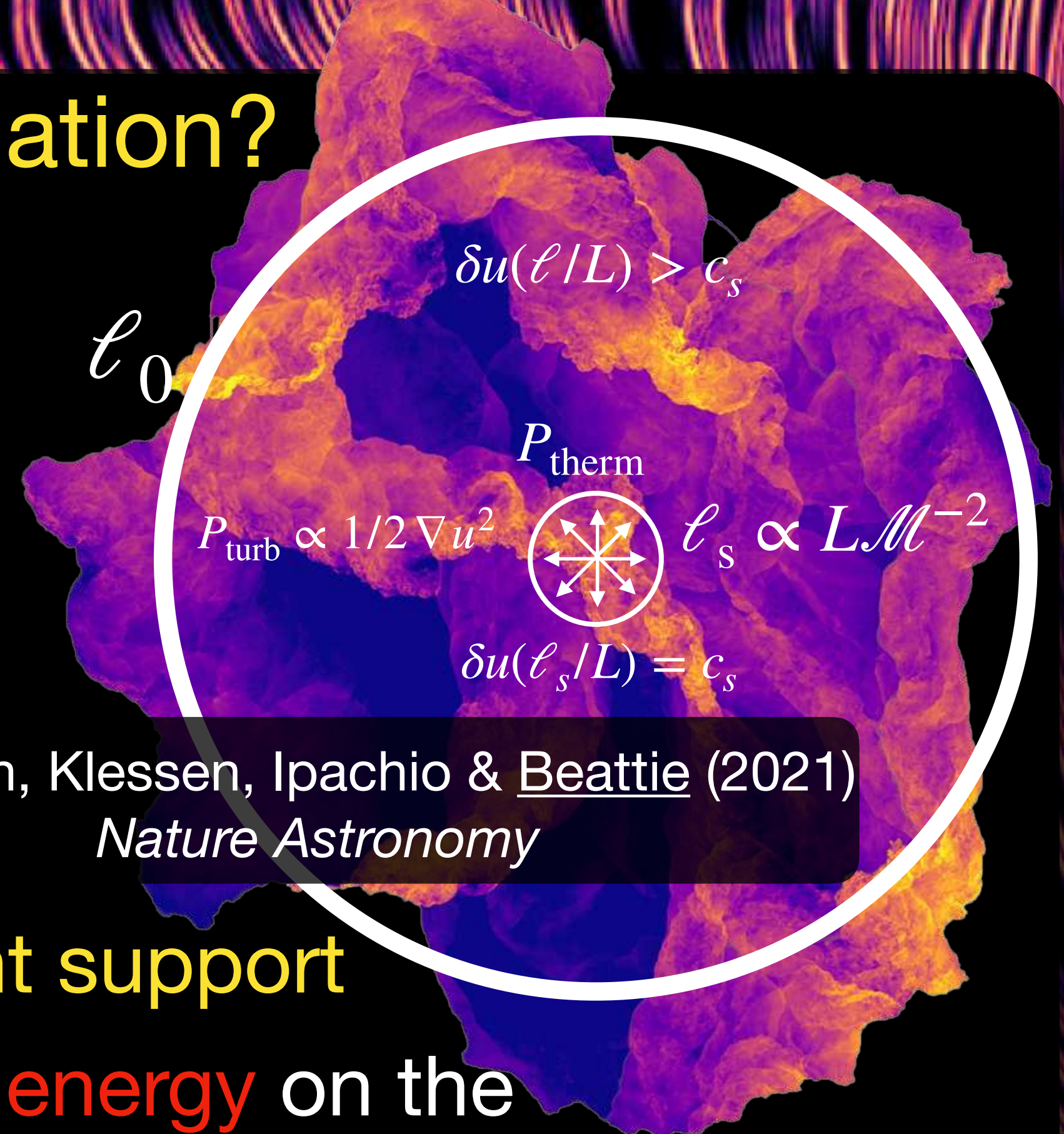
(2012);
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 022);
 el+(2022);

Why should astrophysicists care — star formation?

Sink particle $\propto$ stellar IMF



Nam, Federrath & Krumholz (2021)



Federrath, Klessen, Ipachio & Beattie (2021)
Nature Astronomy

Turbulent support

More energy on the small-scales (more small-scale support) implies less high-mass stars formation

$$\mathbf{u} \cdot \nabla \otimes \mathbf{u} = \frac{1}{2} \nabla u^2 - \mathbf{u} \times \nabla \times \mathbf{u}$$

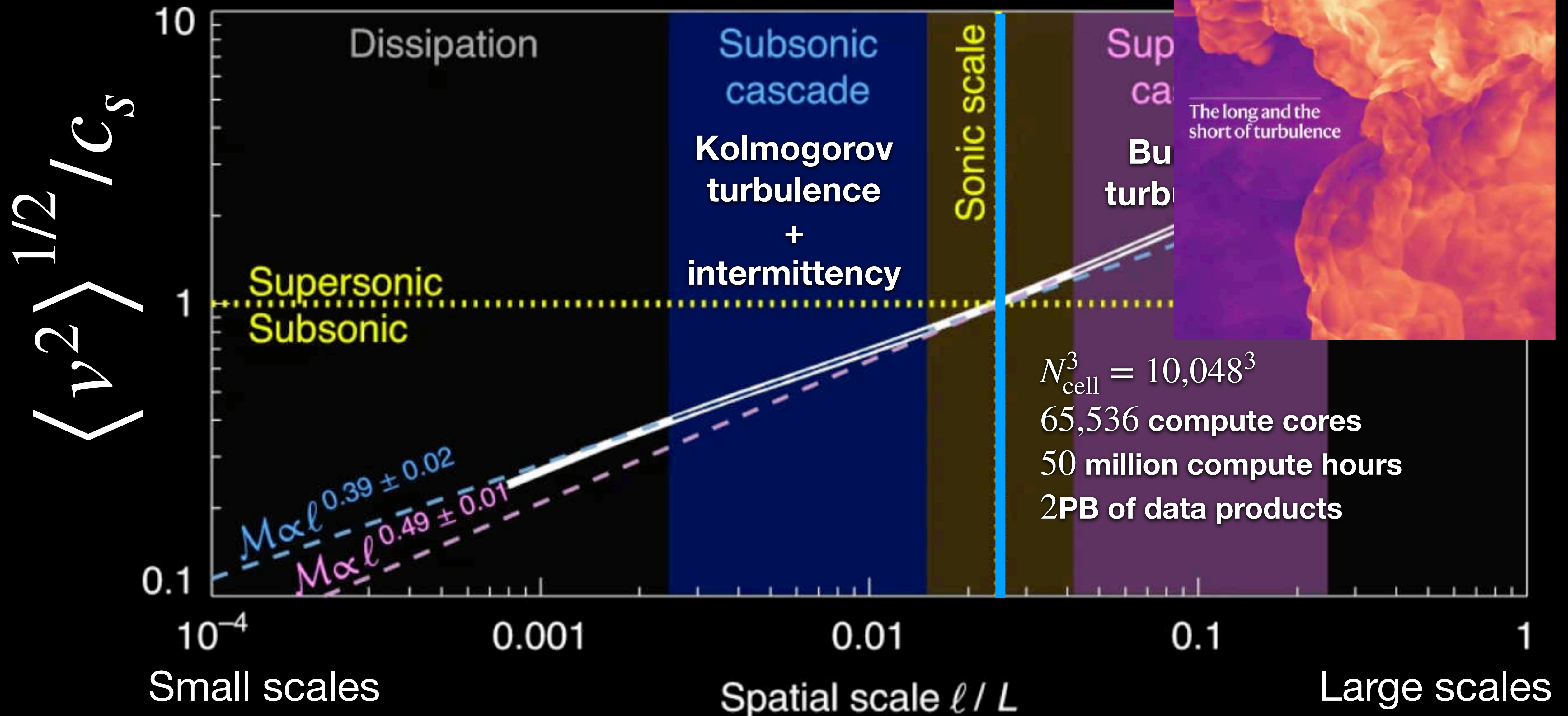
Magnetic reconnection in a turbulent plasma (visualisation by J. Beattie).

The energy cascade in supersonic HD turbulence

Federrath, Klessen, Ipachio & Beattie (2021)

Nature Astronomy

$$\ell_s \sim (0.1 \text{ pc})^{-1}$$

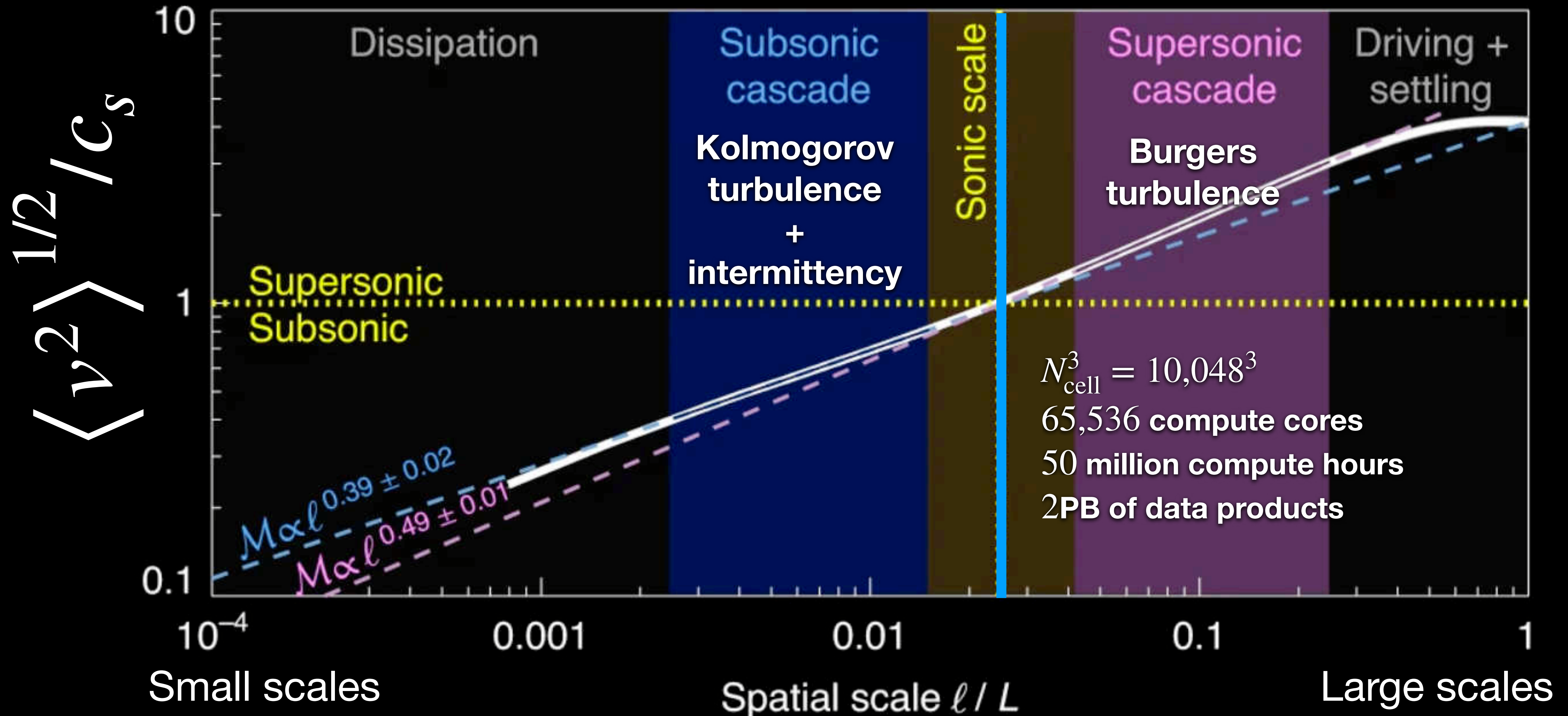


The energy cascade in supersonic HD turbulence

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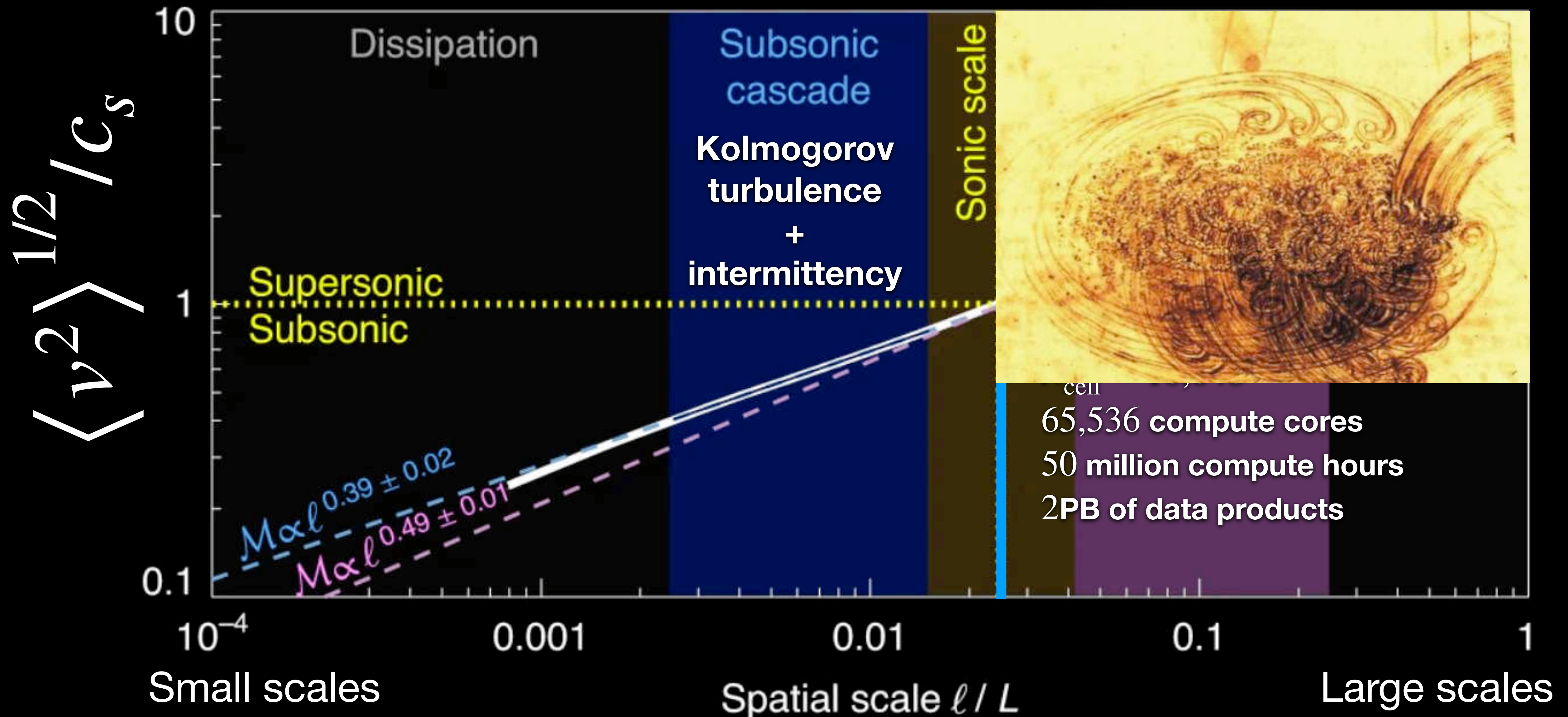
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Federrath, Klessen, Ipachio & Beattie (2021)
Nature Astronomy

$$\mathcal{E}(k) \sim k^{-5/3}$$

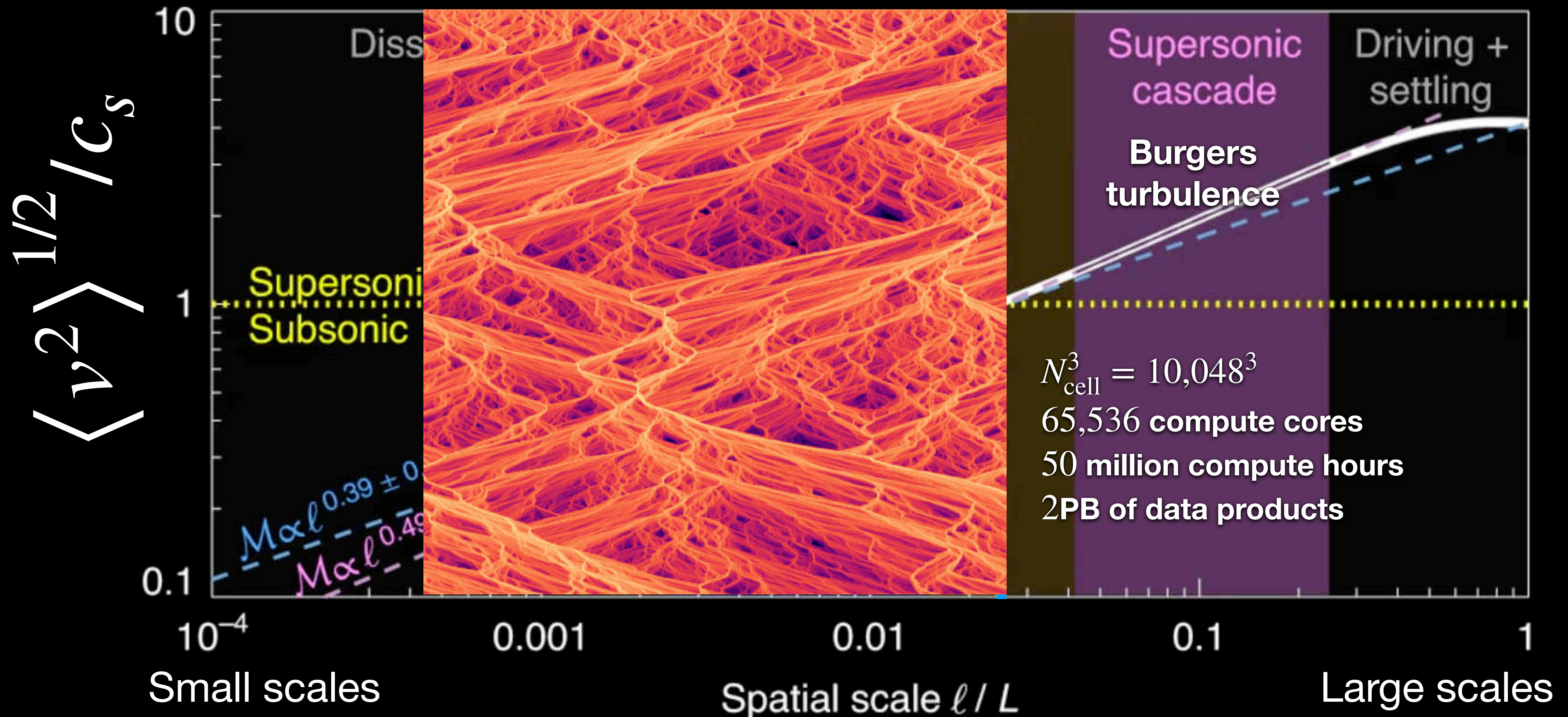


The energy cascade in supersonic HD turbulence

Federrath, Klessen, Ipachio & Beattie (2021)

Nature Astronomy

$$\mathcal{E}(k) \sim k^{-2}$$



The energy cascade in supersonic HD turbulence

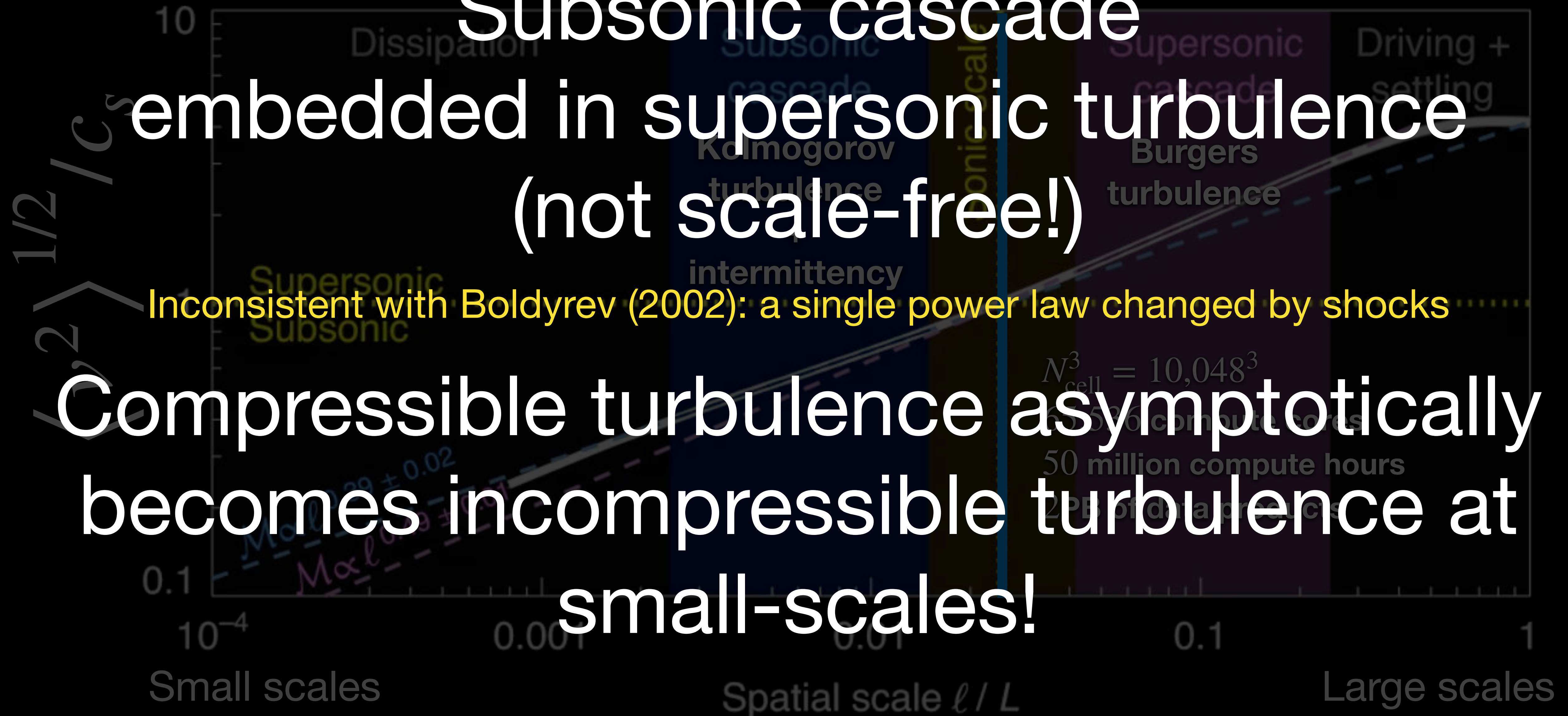
Federrath, Klessen, Ipachio & Beattie (2021)

Nature Astronomy

Subsonic cascade
embedded in supersonic turbulence
(not scale-free!)

Inconsistent with Boldyrev (2002): a single power law changed by shocks

Compressible turbulence asymptotically
becomes incompressible turbulence at
small-scales!

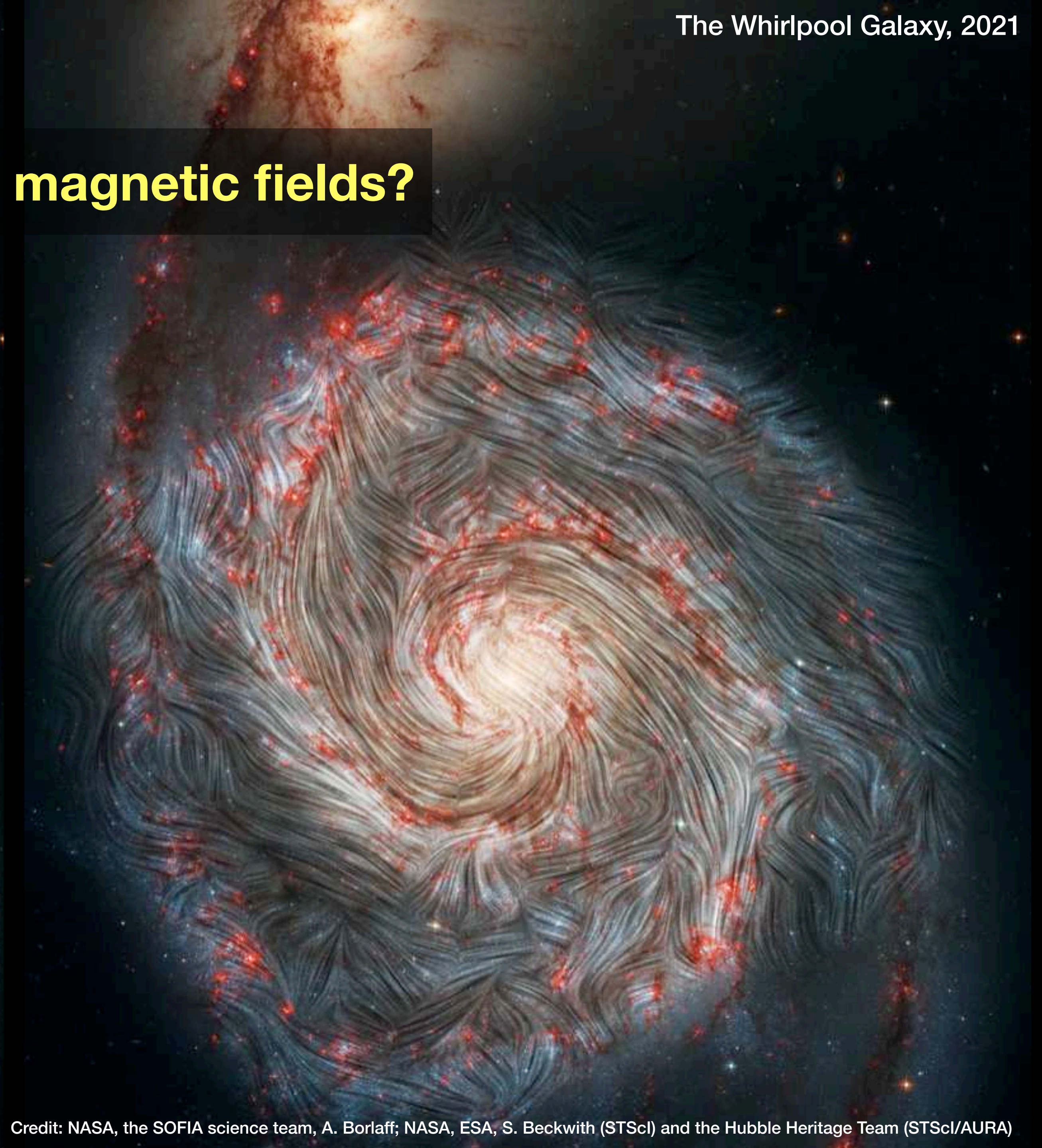
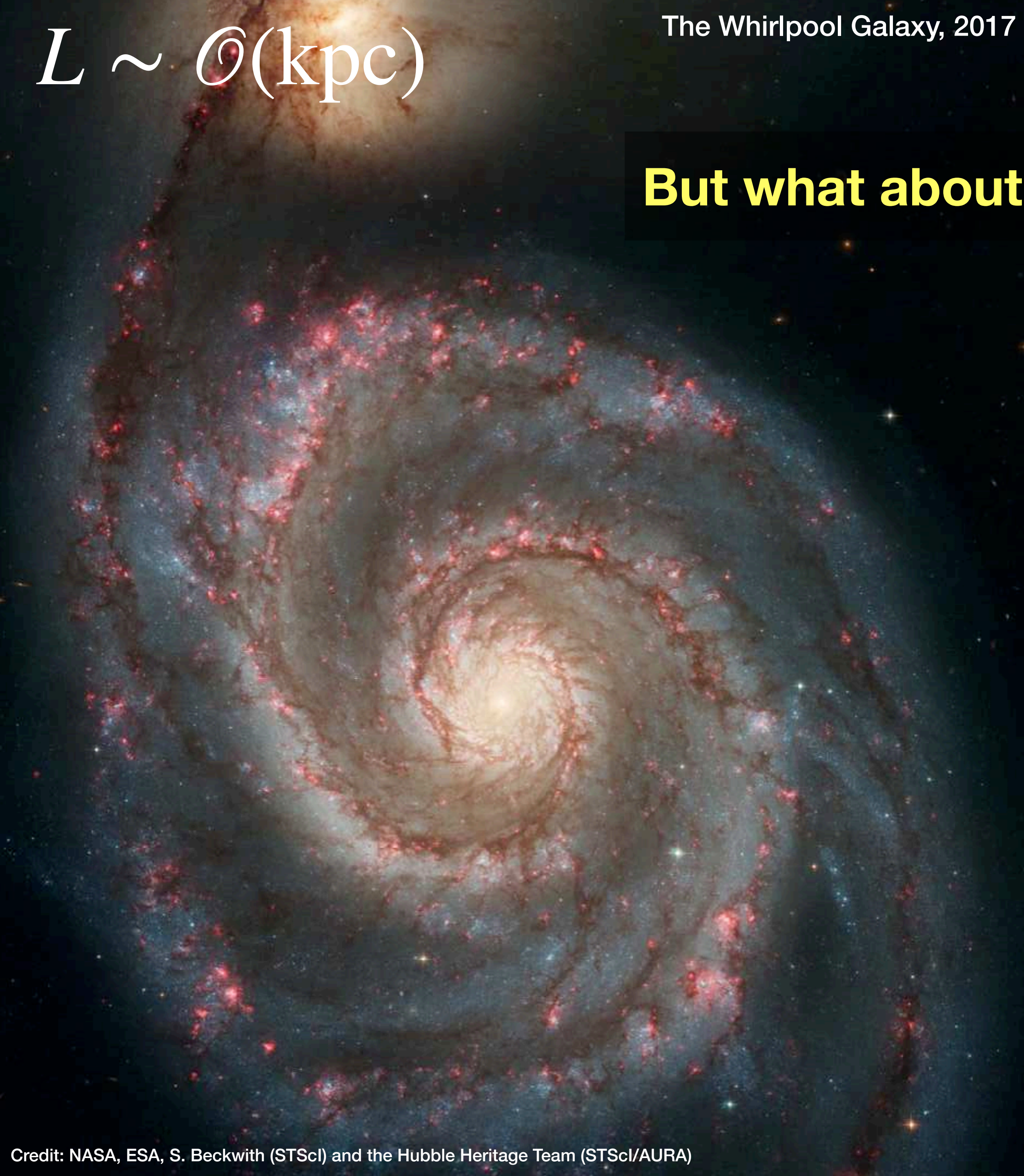


$$L \sim \mathcal{O}(\text{kpc})$$

The Whirlpool Galaxy, 2017

The Whirlpool Galaxy, 2021

But what about magnetic fields?



$$L \sim \mathcal{O}(\text{pc})$$

Soler, 2019, A&A

Irreducible — add magnetic flux to the cascade on all scales

straight, strong fields ($\kappa \propto B^{-1/2}$; Schekochihin+2004) penetrating through Orion A
Orion A in infrared; ESA/Herschel/Planck; J. D. Soler, MPIA

10,080³ magnetized supersonic turbulence simulation (cold ISM analogue)

Beattie, Federrath, Klessen, Cielo & Bhattacharjee

1. What is the scaling of the energy cascade in compressible MHD turbulence?
2. How are the characteristic scales organized in the compressible interstellar medium turbulence?
3. What are the saturation physics of the turbulent dynamo?

PI of a three total 230million core-hour projects on superMUC-NG

ILES of compressible MHD turbulence

Turbulence: $\sigma_V/c_s \approx 4$, $\ell_0 = L/2$

Magnetic fields: $B = b_{\text{turb}}$, $\mathcal{M}_A \approx 2$

Three experiments for convergence tests:

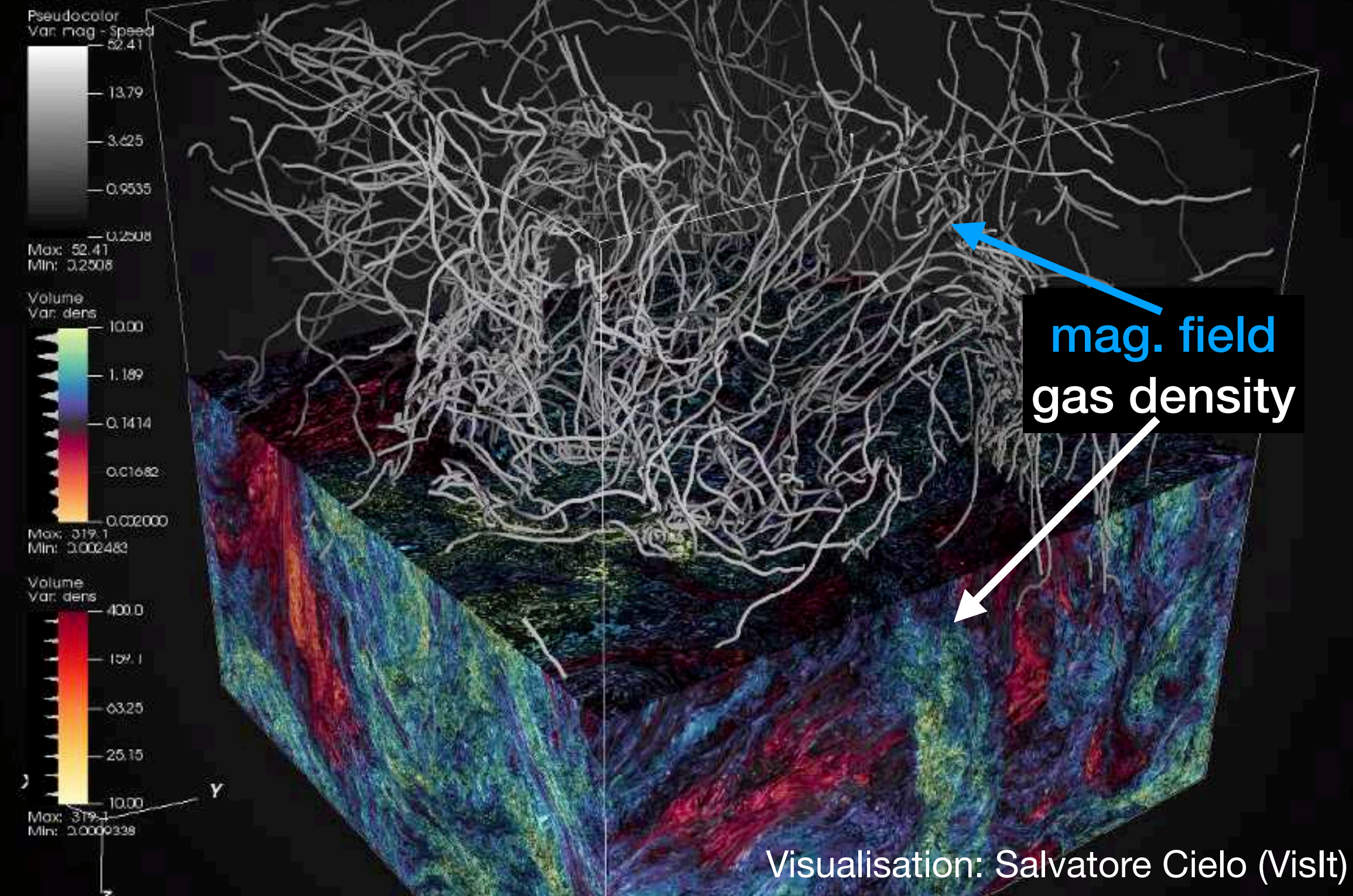
- **LOW-RES:** 2,520³ (0.3Mcore-h, 8,640cores)
- **MID-RES:** 5,040³ (4.0Mcore-h, 34,560cores)
- **HIGH-RES:** 10,080³ (80.0Mcore-h, 148,240cores)

3.45PB in data products $R_m \sim Re \gtrsim 10^6$, $P_m \sim 1 - 2$

Broad Code details:

- Highly-modified version of finite volume code *FLASH*, second-order in space approximate Riemann (PPM) solver with framework outlined in Bouchut+2010, tested in *FLASH* in Waagen+2011.
- Ideal (ILES) compressible non-helical, isothermal MHD turbulence with finite correlation time (OU process; Federrath+2022).

DB: EXTREME_Turb_hdf5_plt_cnt_0100
Cycle: 118366 Time: 1.25



10,080³ magnetized supersonic turbulence simulation (cold ISM analogue)

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Bhattacharjee

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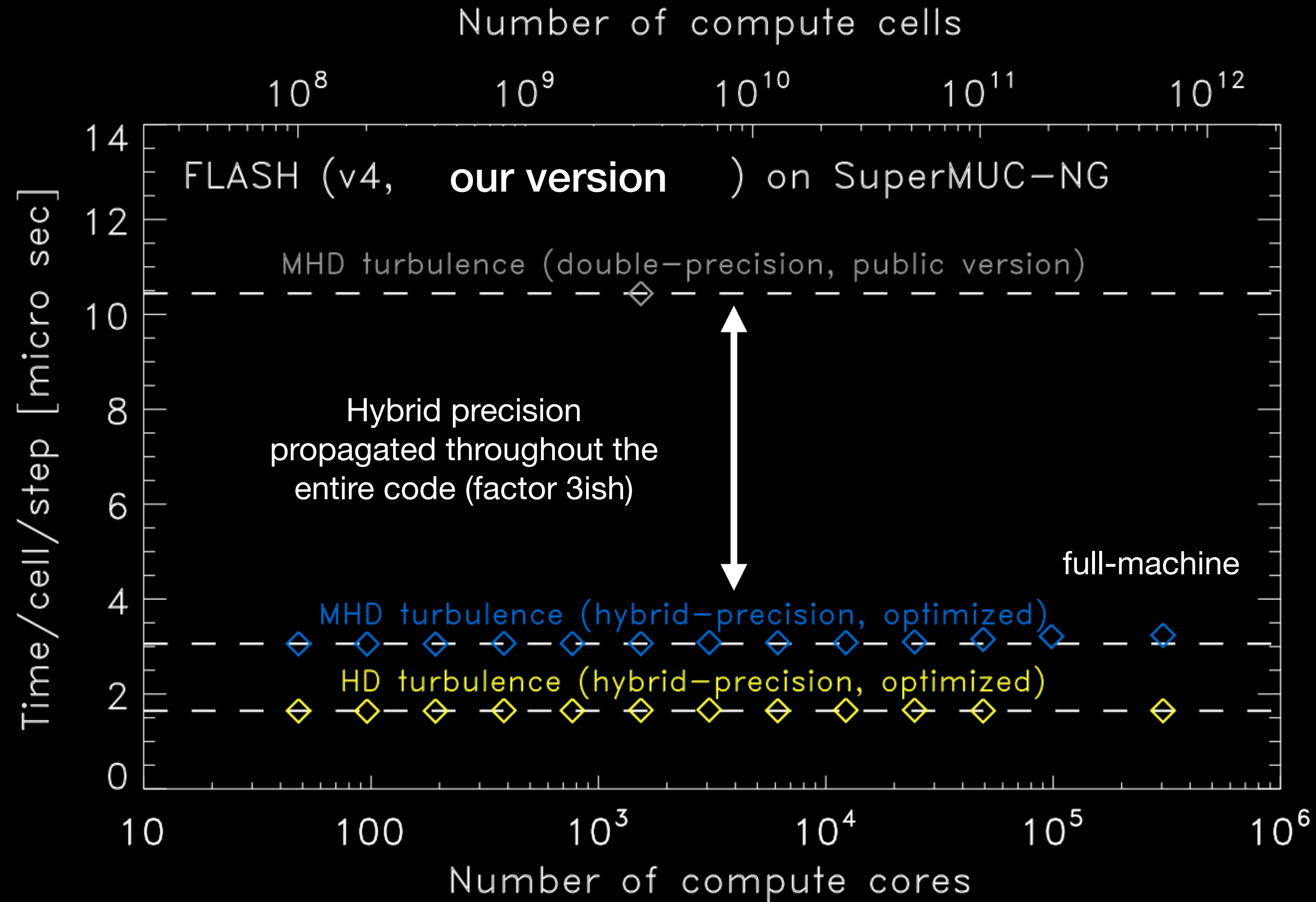
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Perfect weak scaling on SuperMUC-NG



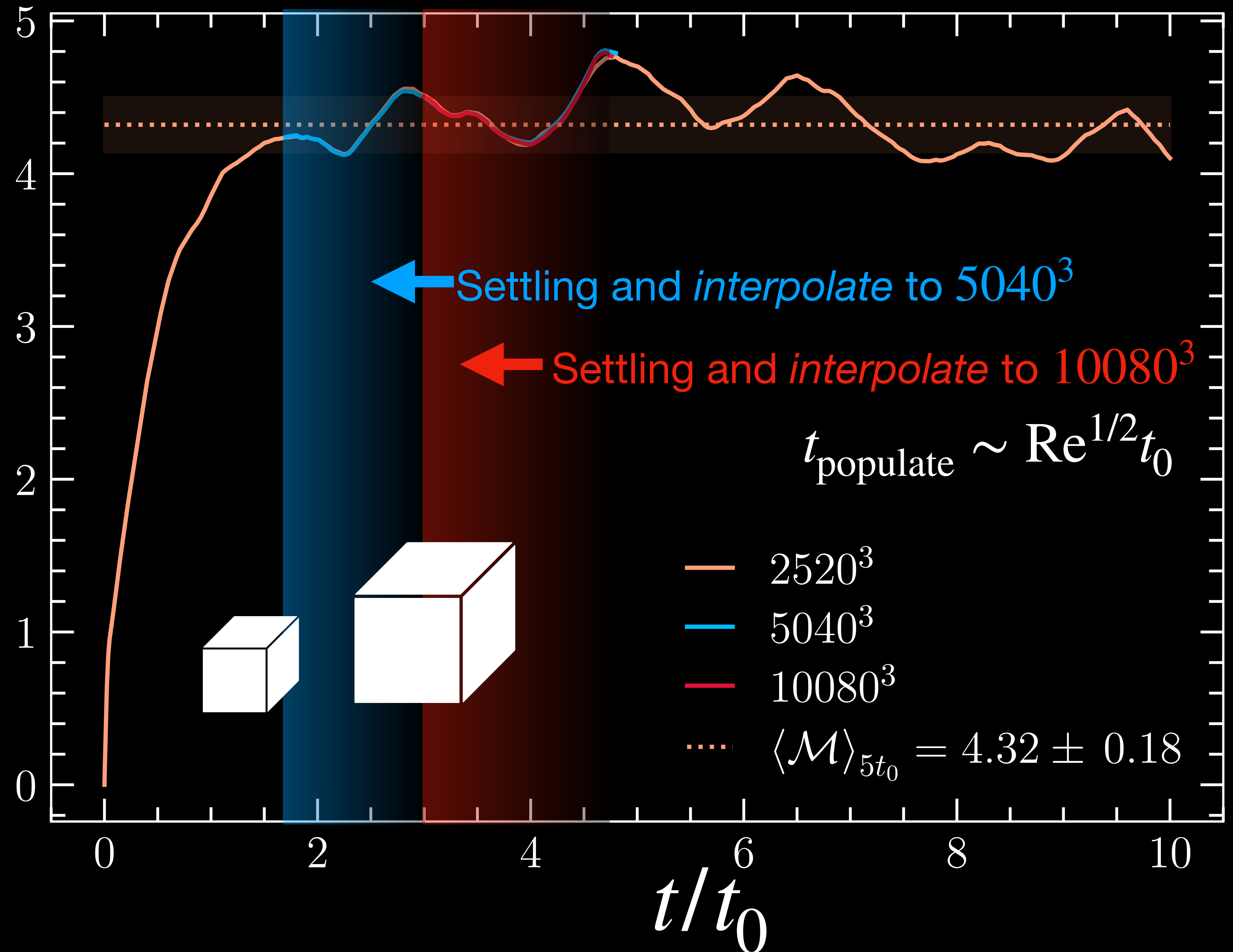
Max: 319.4
Min: 2.0000338

Visualisation: Salvatore Cielo (VisIt)

Volume integral Quantities

$$\mathcal{M} = \left\langle \frac{u^2}{c_s^2} \right\rangle_{\mathcal{V}}^{1/2}$$

Interpolation to
generate ICs for
successively higher
resolution
experiments

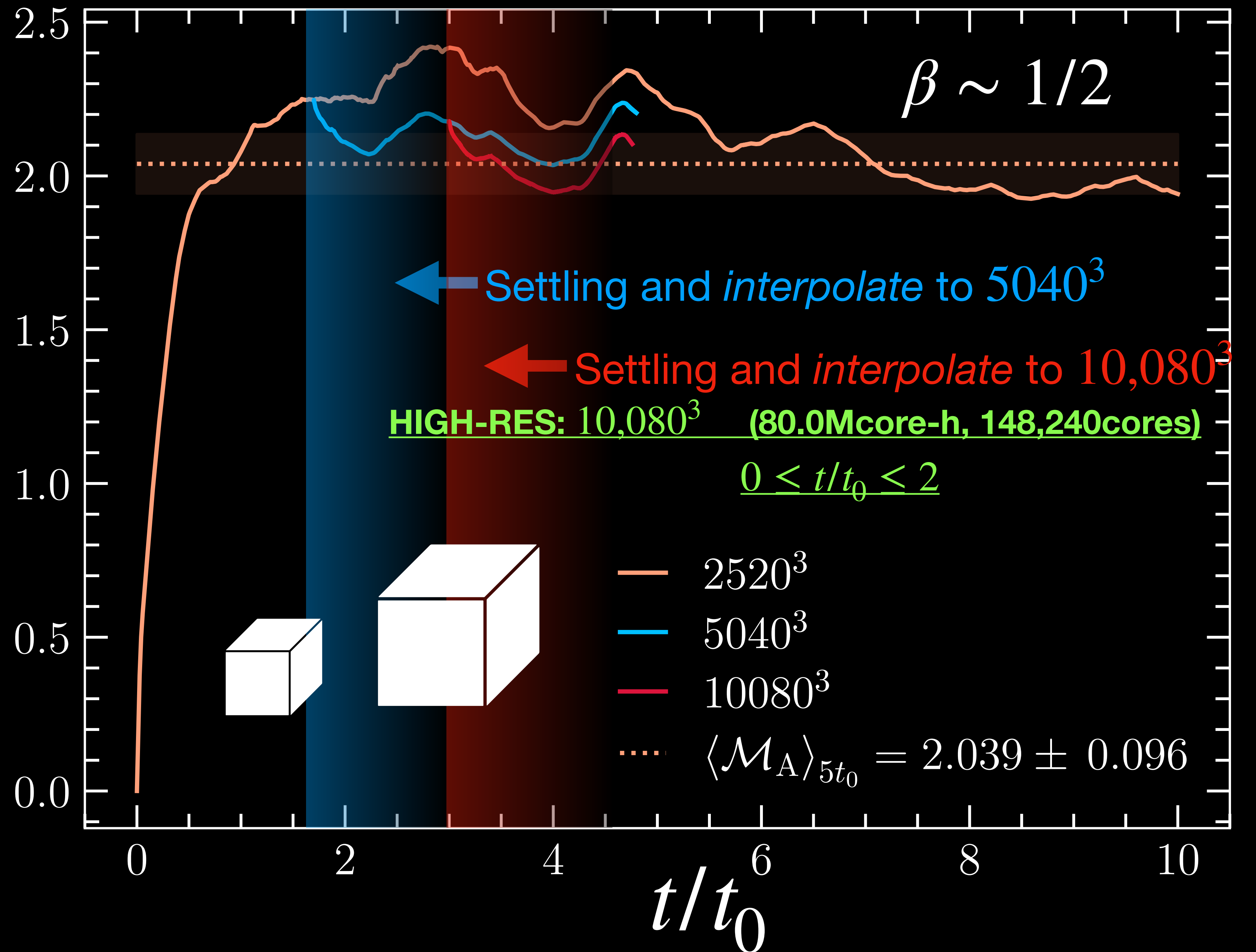


Volume integral Quantities

$$\mathcal{M}_A = \sqrt{\frac{E_{\text{kin}}}{E_{\text{mag}}}}$$

Interpolation to
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resolution
experiments

$\mathcal{M}_A$

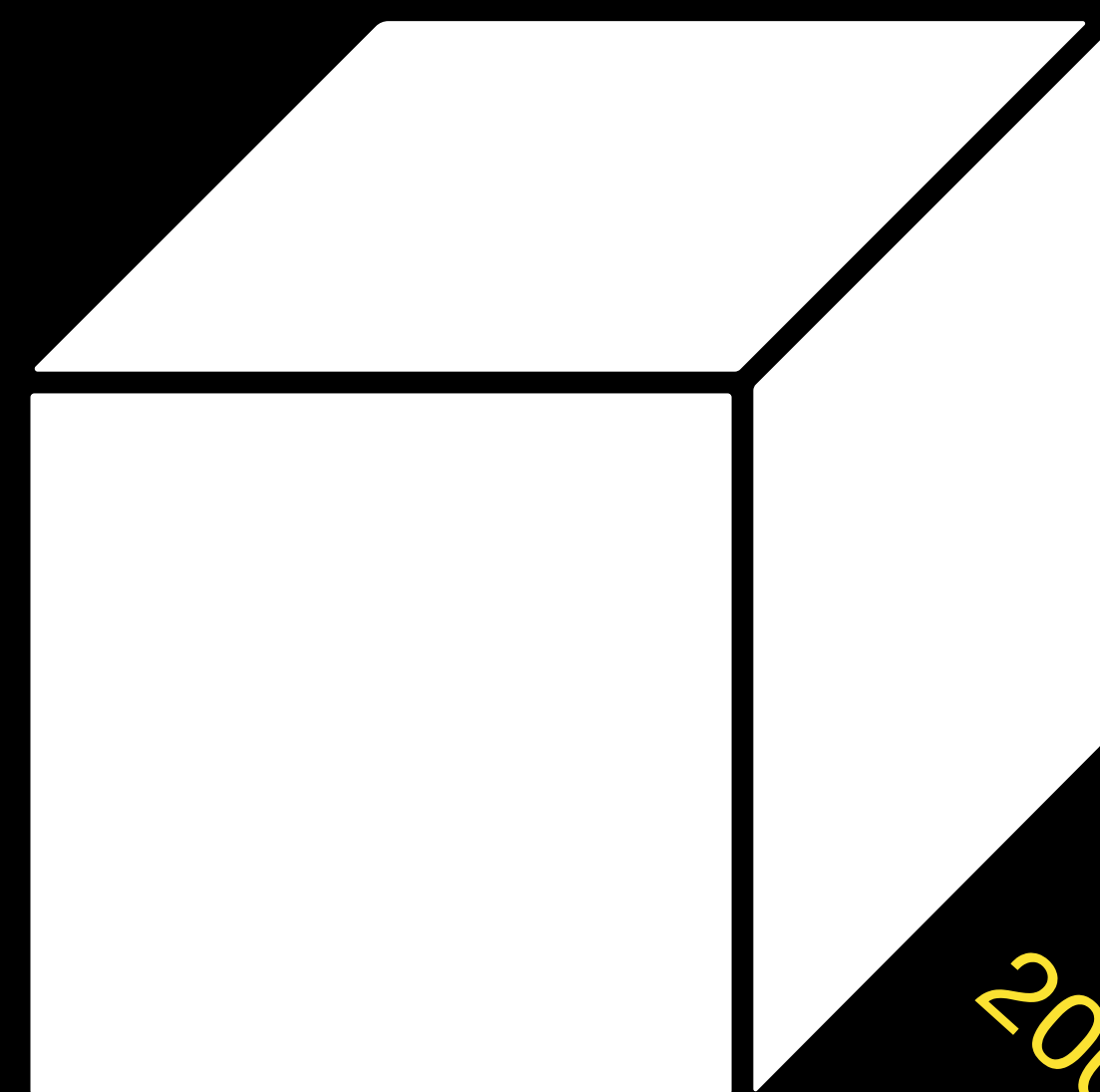


compute disk

Data dumps at 10x per correlation time for averages across the statistically steady state

Read out domain into HDF5 chunks, grouped by m nodes writing each chunk

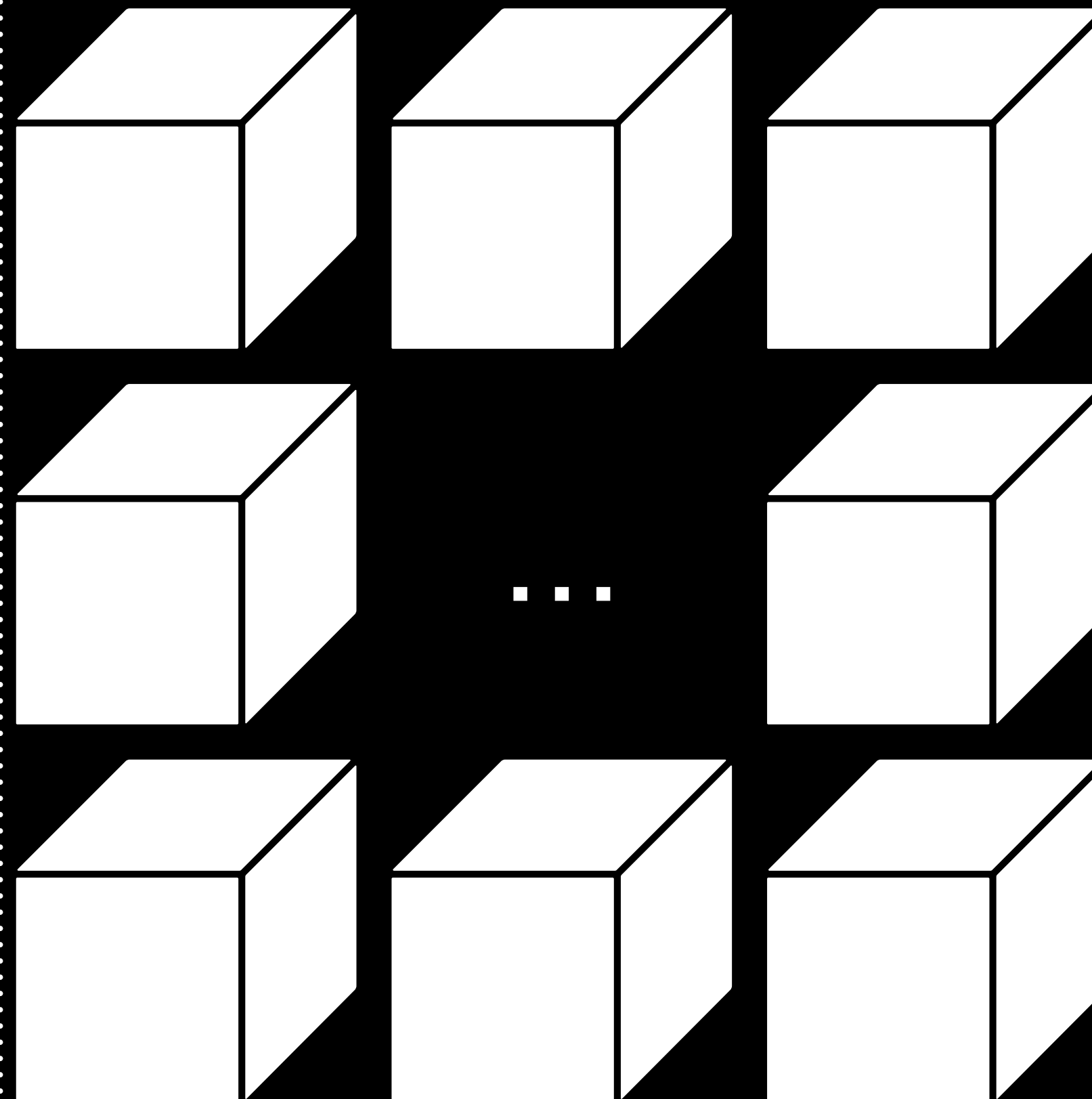
Merge into single HDF5 file in post-processing (C++ mpi tool)



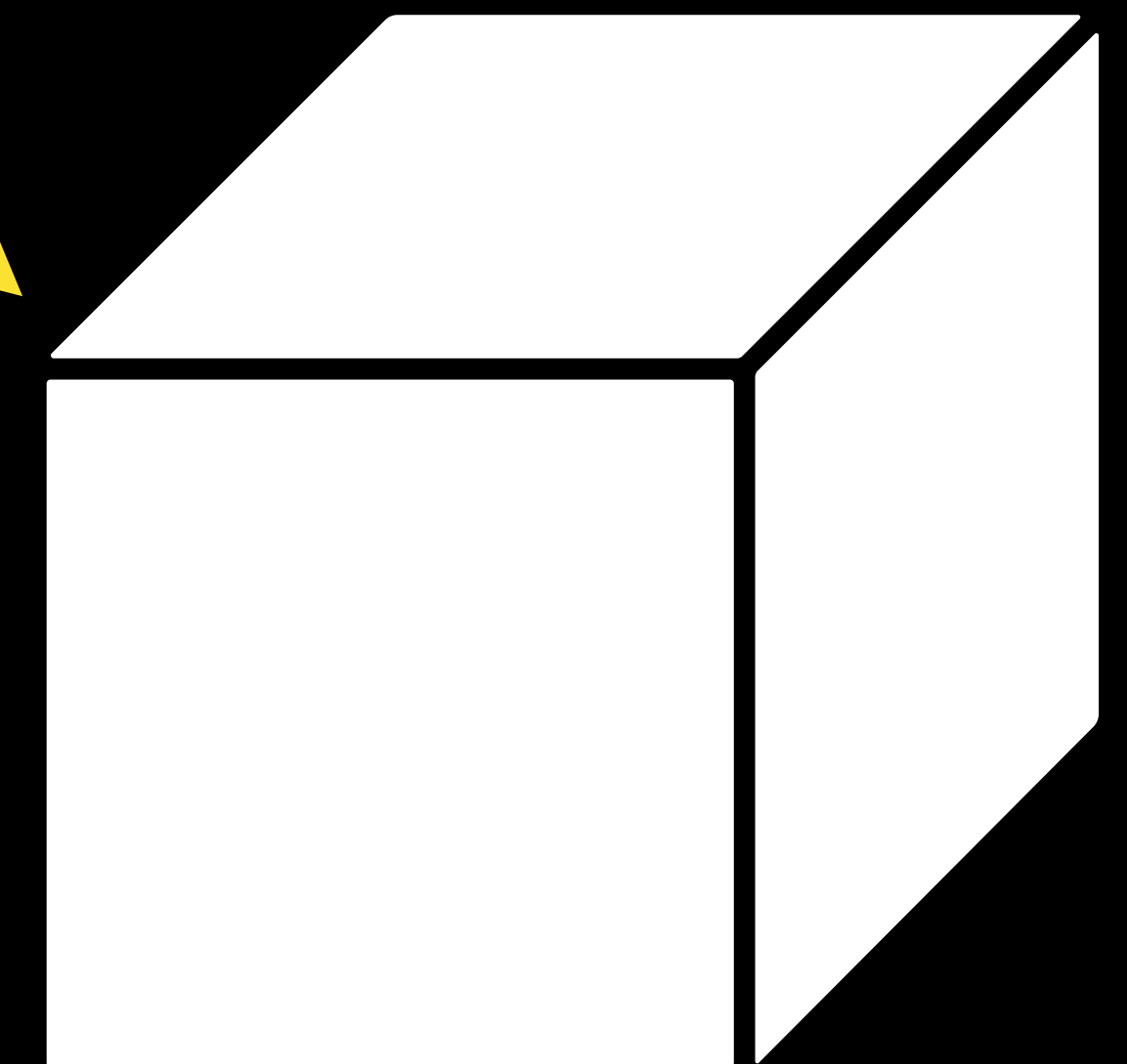
$10,080^3$

200 GB/sec

Almost max. bandwidth for SuperMUC-NG scratch



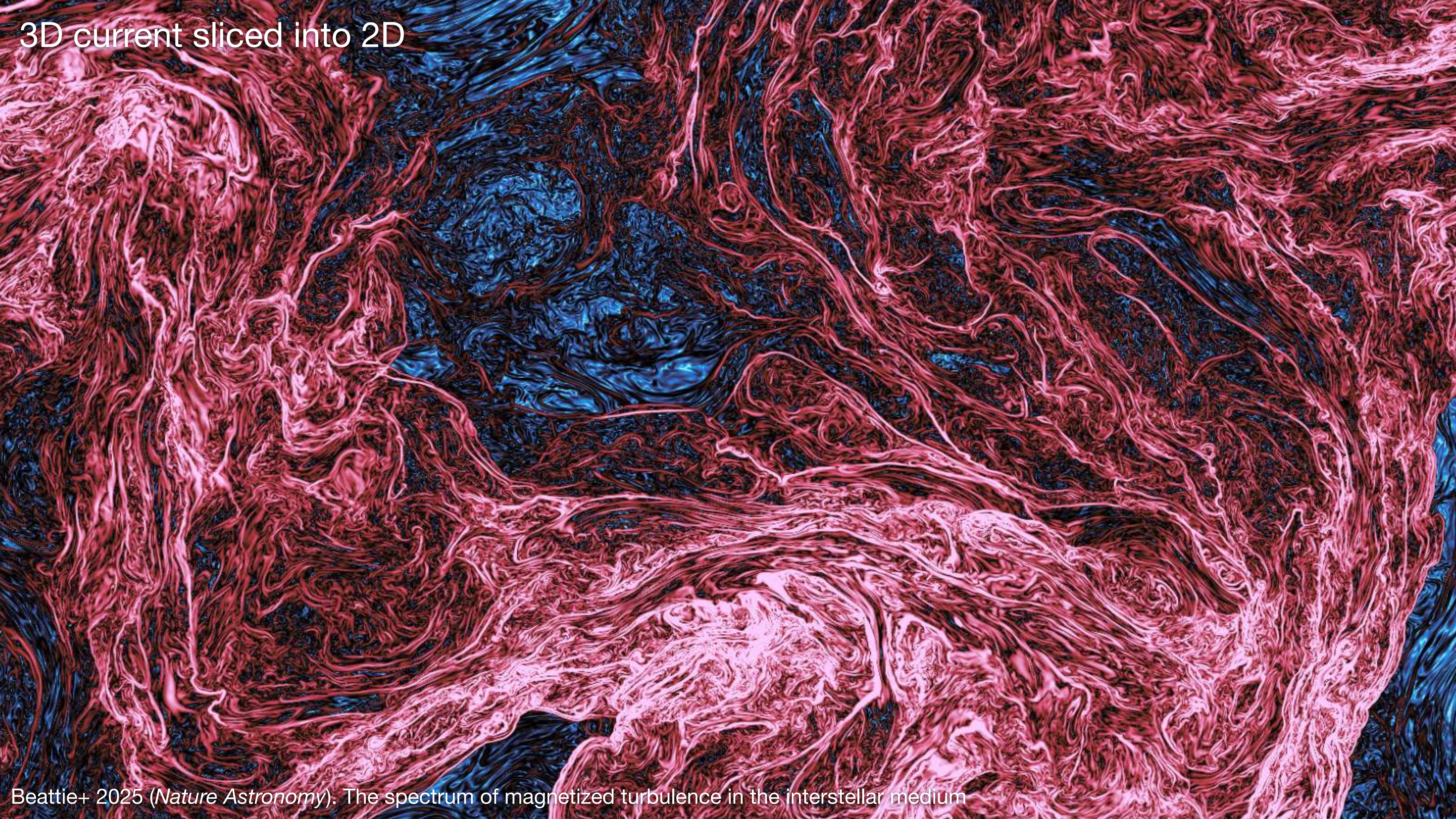
180 sec to write a 28 TB checkpoint file



$10,080^3$

C++ & FFTW backend
Power spectrum on 50,000 cores.
Spectra in ~ 3mins

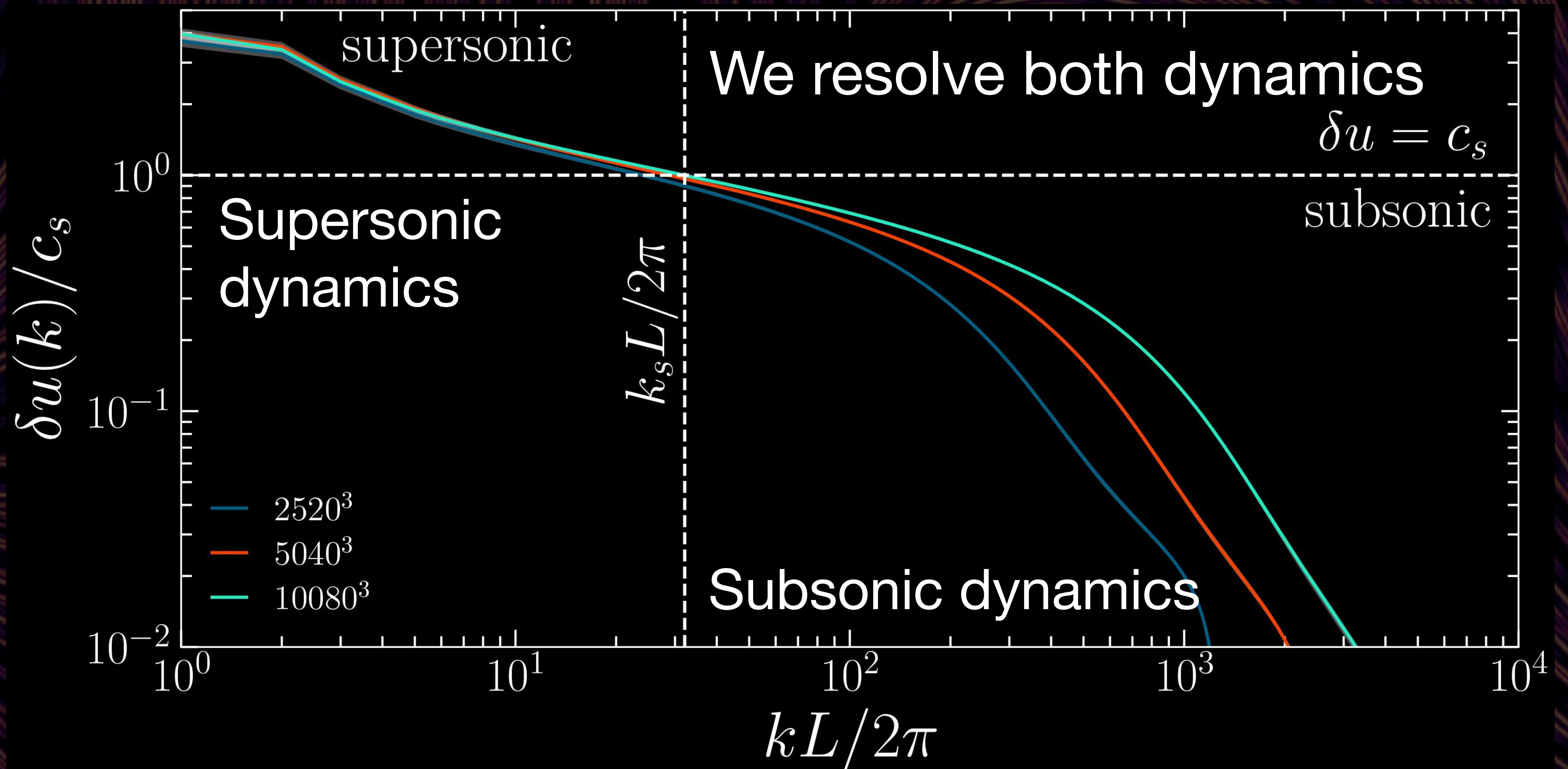




3D current sliced into 2D

Beattie+ 2025 (*Nature Astronomy*). The spectrum of magnetized turbulence in the interstellar medium

Two important scales — the sonic scale



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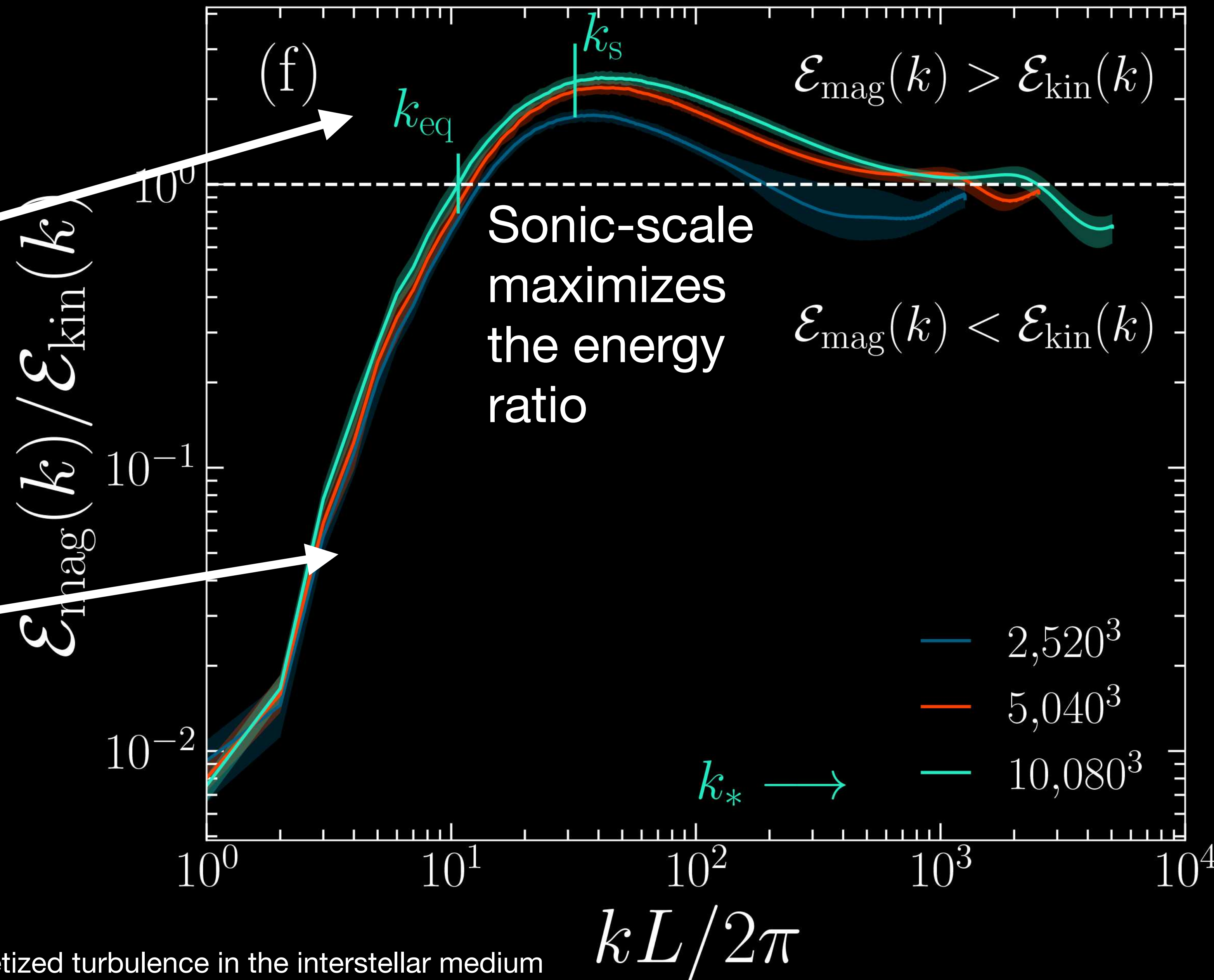
Two important scales — the Alfvén scale

sub-Alfvénic

k_{eq}

super-Alfvénic

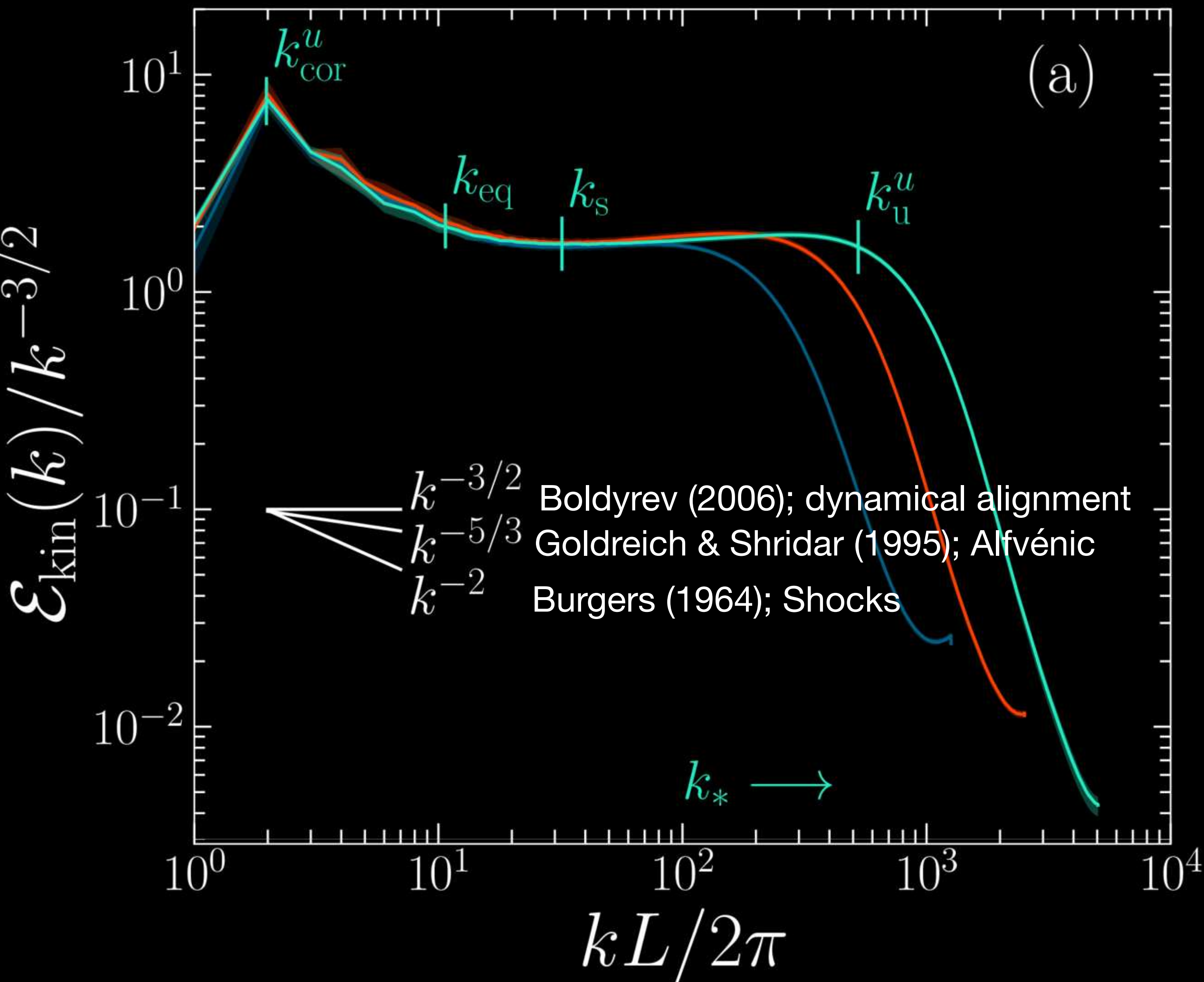
Multiple regimes
in a single domain!



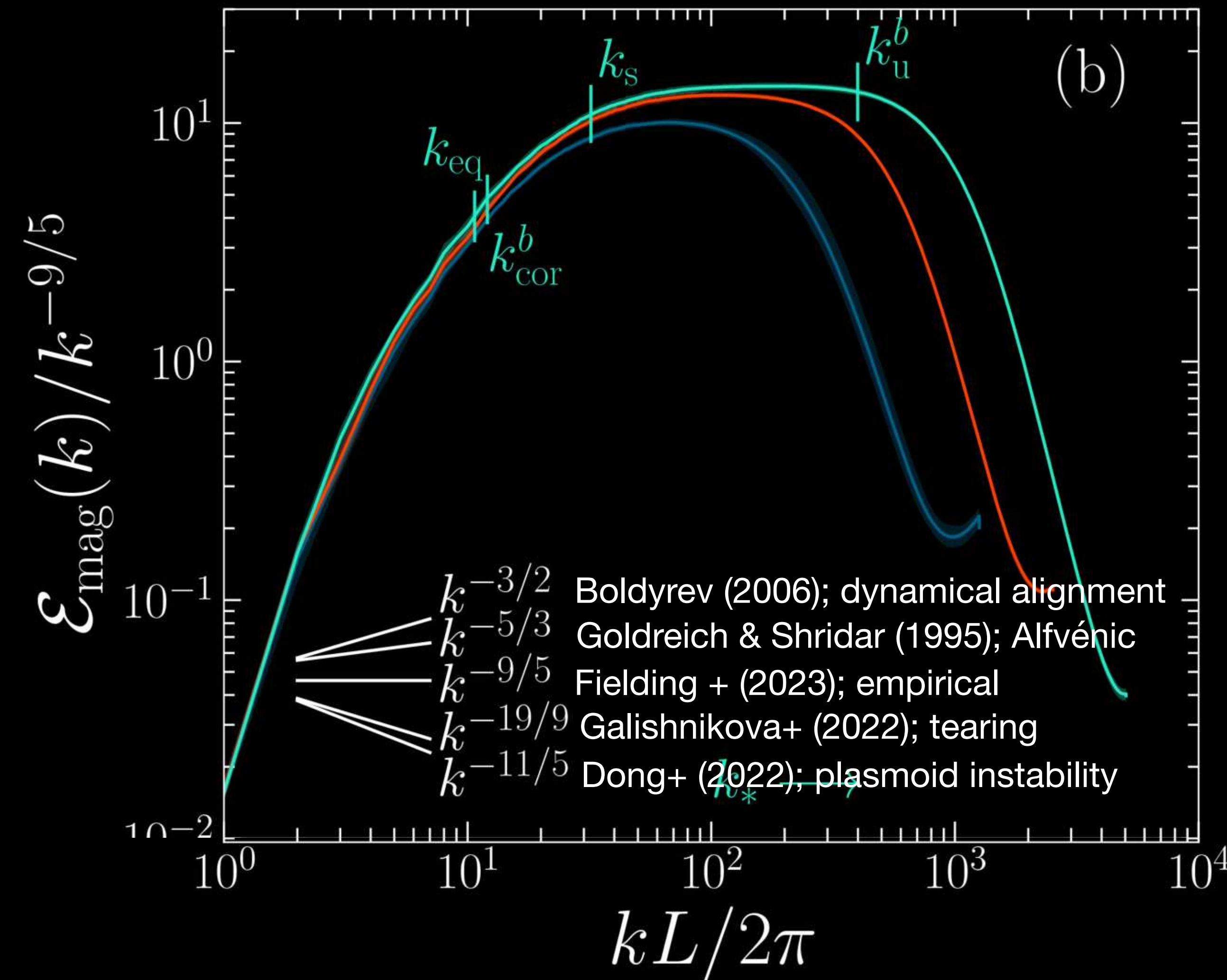
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Energy spectra: the disparate lives of u and b fluctuations

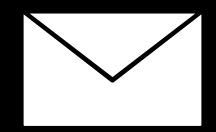


Two power laws



One power law, only
emerging at very high Rm

Thanks, questions?



james.beattie@princeton.edu



[@astro_magnetism](https://twitter.com/astro_magnetism)

Not enough time to talk about many more key results

- Ideas about the origin of alignment
- Other primitive variable / curl alignment structure function
- In situ current sheet and plasmoid development
- Comparison of density fluctuations with MMS data
- Turbulent dynamo saturation through alignment
- Comparisons with more global supernova-driven turbulence

The future is bright

- Next generation of $10,000^3$ simulations with large-scale field integrating now
- Moving to also have AthenaK capabilities to do GPU counterpart runs
- Pushing towards $20,000^3$ (500TB 3D dumps) to further explore asymptotic state